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Bank database acid properties and isolation

Program:

```
1  ● CREATE DATABASE Bank_ACID_DB;
2  USE Bank_ACID_DB;
3
4  ○ CREATE TABLE Customer (
5      Customer_ID INT PRIMARY KEY,
6      Customer_Name VARCHAR(100) NOT NULL,
7      Phone VARCHAR(15),
8      City VARCHAR(50)
9  );
10
11 ● ○ CREATE TABLE Account (
12     Account_No INT PRIMARY KEY,
13     Customer_ID INT,
14     Account_Type VARCHAR(20),
15     Balance DECIMAL(12,2),
16     Branch VARCHAR(50),
17     FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
18 );
19
20 ● ○ CREATE TABLE Bank_Transaction {
21     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
22     Account_No INT,
23     Transaction_Type VARCHAR(20),
24     Amount DECIMAL(12,2),
25     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
26     FOREIGN KEY (Account_No) REFERENCES Account(Account_No)
27 };
```

Output:

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
1	10:35:22	CREATE DATABASE Bank_ACID_DB	1 row(s) affected	0.000 sec
2	10:35:24	USE Bank_ACID_DB	0 row(s) affected	0.000 sec
3	10:35:25	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	0 row(s) affected	0.031 sec
4	10:35:28	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARC...	0 row(s) affected	0.063 sec
5	10:35:34	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account...	0 row(s) affected	0.078 sec

Program:

```
29 ● INSERT INTO Customer
30 (Customer_ID, Customer_Name, Phone, City)
31 VALUES
32 (101, 'Tejaswini', '9876543210', 'Hyderabad'),
33 (102, 'Priya Sharma', '9876543211', 'Vijayawada'),
34 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
35 (104, 'Sneha Rao', '9876543213', 'Chennai'),
36 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
37
38 ● INSERT INTO Account
39 (Account_No, Customer_ID, Account_Type, Balance, Branch)
40 VALUES
41 (10001, 101, 'Savings', 50000, 'Hyderabad'),
42 (10002, 102, 'Savings', 75000, 'Vijayawada'),
43 (10003, 103, 'Current', 120000, 'Bangalore'),
44 (10004, 104, 'Savings', 45000, 'Chennai'),
45 (10005, 105, 'Current', 90000, 'Hyderabad');
46
47 ● INSERT INTO Bank_Transaction
48 (Account_No, Transaction_Type, Amount)
49 VALUES
50 (10001, 'DEPOSIT', 10000),
51 (10002, 'DEPOSIT', 15000),
52 (10003, 'WITHDRAW', 20000),
53 (10004, 'DEPOSIT', 5000),
54 (10005, 'WITHDRAW', 10000);
55
56 ● SELECT * FROM Customer;
```

Output:

The screenshot displays a database management interface. The top section features a 'Result Grid' with a table of customer data. The table has columns for Customer_ID, Customer_Name, Phone, and City. The data includes five rows with IDs 101 through 105, and a final row with NULL values. To the right of the table is a sidebar with icons for 'Result Grid', 'Form Editor', and 'Field Types'. Below the table are buttons for 'Apply' and 'Revert', and links for 'Context Help' and 'Snippets'. The bottom section, titled 'Output', shows a log of database actions. The log includes a table with columns for #, Time, Action, Message, and Duration / Fetch. The actions listed are: creating a table, inserting data into the Customer table, inserting data into the Account table, inserting data into the Bank_Transaction table, and two SELECT queries on the Customer table.

	Customer_ID	Customer_Name	Phone	City
▶	101	Tejaswini	9876543210	Hyderabad
	102	Priya Sharma	9876543211	Vijayawada
	103	Arjun Reddy	9876543212	Bangalore
	104	Sneha Rao	9876543213	Chennai
	105	Kiran Kumar	9876543214	Hyderabad
*	NULL	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
✓ 5	10:35:34	CREATE TABLE Bank_Transaction (Transa...	0 row(s) affected	0.078 sec
✓ 6	10:36:31	INSERT INTO Customer (Customer_ID,Custome...	5 row(s) affected Records: 5 Duplicates: 0 Wa...	0.000 sec
✓ 7	10:36:34	INSERT INTO Account (Account_No,Customer...	5 row(s) affected Records: 5 Duplicates: 0 Wa...	0.015 sec
✓ 8	10:36:37	INSERT INTO Bank_Transaction (Account_No...	5 row(s) affected Records: 5 Duplicates: 0 Wa...	0.015 sec
✓ 9	10:36:40	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓ 10	10:37:11	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Program:

```
57 ● SELECT * FROM Account;
58 ● SELECT * FROM Bank_Transaction;
59
60 ● START TRANSACTION;
61
62 ● UPDATE Account
63   SET Balance=Balance+5000
64   WHERE Account_No=10001;
65
66 ● SELECT *
67   FROM Account
68   WHERE Account_No=10001;
69
70 ● COMMIT;
71
72 ● SELECT *
73   FROM Account
74   WHERE Account_No=10001;
75
76 ● START TRANSACTION;
77
78 ● UPDATE Account
79   SET Balance=Balance-10000
80   WHERE Account_No=10001;
81
82 ● SELECT *
83   FROM Account
84   WHERE Account_No=10001;
85
```

Output:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
●	NULL	NULL	NULL	NULL	NULL

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_D
▶	1	10001	DEPOSIT	10000.00	2026-08-16 10:
	2	10002	DEPOSIT	15000.00	2026-08-16 10:
	3	10003	WITHDRAW	20000.00	2026-08-16 10:
	4	10004	DEPOSIT	5000.00	2026-08-16 10:
	5	10005	WITHDRAW	10000.00	2026-08-16 10:
★	NULL	NULL	NULL	NULL	NULL

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
★	NULL	NULL	NULL	NULL	NULL

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
★	NULL	NULL	NULL	NULL	NULL

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
★	NULL	NULL	NULL	NULL	NULL

✓	11	10:38:51	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓	12	10:39:05	SELECT * FROM Bank_Transaction LIMIT 0, 1...	5 row(s) returned	0.000 sec / 0.000 sec
✓	13	10:39:20	START TRANSACTION	0 row(s) affected	0.000 sec
✓	14	10:39:23	UPDATE Account SET Balance=Balance+500...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	15	10:39:25	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.000 sec / 0.000 sec
✓	16	10:39:42	COMMIT	0 row(s) affected	0.015 sec

✓	17	10:39:44	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.000 sec / 0.000 sec
✓	18	10:48:14	START TRANSACTION	0 row(s) affected	0.000 sec
✓	19	10:48:17	UPDATE Account SET Balance=Balance-1000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	20	10:48:24	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.000 sec / 0.000 sec

Program:

```
114 ● SELECT * FROM Account;
115 ● START TRANSACTION;
116 ● UPDATE Account
117     SET Balance=Balance-10000
118     WHERE Account_No=10001;
119 ● UPDATE Account
120     SET Balance=Balance+10000
121     WHERE Account_No=10002;
122 ● SELECT *
123     FROM Account
124     WHERE Account_No IN (10001,10002);
125 ● COMMIT;
126 ● SELECT Account_No,Balance
127     FROM Account
128     WHERE Account_No IN (10001,10002);
129 ● START TRANSACTION;
130 ● UPDATE Account
131     SET Balance=Balance-20000
132     WHERE Account_No=10001;
133 ● UPDATE Account
134     SET Balance=Balance+20000
135     WHERE Account_No=10002;
136 ● ROLLBACK;
137 ● SELECT *
138     FROM Account
139     WHERE Account_No IN (10001,10002);
140 ● START TRANSACTION;
```

Output:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
●	NULL	NULL	NULL	NULL	NULL

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	35000.00	Hyderabad
	10002	102	Savings	85000.00	Vijayawada
●	NULL	NULL	NULL	NULL	NULL

✓	21	10:50:30	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓	22	10:50:45	START TRANSACTION	0 row(s) affected	0.000 sec
✓	23	10:50:47	UPDATE Account SET Balance=Balance-1000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.016 sec
✓	24	10:50:51	UPDATE Account SET Balance=Balance+100...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	25	10:50:53	SELECT * FROM Account WHERE Account_N...	2 row(s) returned	0.000 sec / 0.000 sec
✓	26	10:51:12	COMMIT	0 row(s) affected	0.000 sec

	Account_No	Balance
▶	10001	35000.00
	10002	85000.00
✱	NULL	NULL

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	35000.00	Hyderabad
	10002	102	Savings	85000.00	Vijayawada
✱	NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

Account 11 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 27	10:51:39	SELECT Account_No,Balance FROM Account ...	2 row(s) returned	0.000 sec / 0.000 sec
✓ 28	10:51:54	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 29	10:51:57	UPDATE Account SET Balance=Balance-2000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓ 30	10:51:59	UPDATE Account SET Balance=Balance+200...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓ 31	10:52:02	ROLLBACK	0 row(s) affected	0.016 sec
✓ 32	10:52:04	SELECT * FROM Account WHERE Account_N...	2 row(s) returned	0.000 sec / 0.000 sec

✓	33	10:52:33	START TRANSACTION	0 row(s) affected	0.000 sec
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Program:

```
141 ● UPDATE Account
142     SET Balance=Balance-10000
143     WHERE Account_No=10001;
144 ● UPDATE Account
145     SET Balance=Balance+10000
146     WHERE Account_No=10002;
147 ● COMMIT;
148 ● START TRANSACTION;
149 ● UPDATE Account
150     SET Balance=Balance+5000
151     WHERE Account_No=10001;
152 ● COMMIT;
153 ● SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED;
154 ● START TRANSACTION;
155 ● SELECT *
156     FROM Account
157     WHERE Account_No=10001;
158 ● COMMIT;
159 ● SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
160 ● START TRANSACTION;
161 ● SELECT *
162     FROM Account
163     WHERE Account_No=10001;
164 ● COMMIT;
165 ● SET SESSION TRANSACTION ISOLATION LEVEL REPEATABLE READ;
166 ● START TRANSACTION;
167 ● SELECT *
168     FROM Account
169     WHERE Account_No=10001;
```

Output:

✓	34	10:53:49	UPDATE Account SET Balance=Balance-1000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	35	10:53:52	UPDATE Account SET Balance=Balance+100...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	36	10:53:54	COMMIT	0 row(s) affected	0.000 sec
✓	37	10:53:56	START TRANSACTION	0 row(s) affected	0.000 sec
✓	38	10:53:57	UPDATE Account SET Balance=Balance+500...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	39	10:54:00	COMMIT	0 row(s) affected	0.016 sec

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	30000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

✓	40	10:54:14	SET SESSION TRANSACTION ISOLATION L...	0 row(s) affected	0.000 sec
✓	41	10:54:17	START TRANSACTION	0 row(s) affected	0.000 sec
✓	42	10:54:19	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.000 sec / 0.000 sec
✓	43	10:54:36	COMMIT	0 row(s) affected	0.000 sec
✓	44	10:54:40	SET SESSION TRANSACTION ISOLATION L...	0 row(s) affected	0.000 sec
✓	45	10:54:43	START TRANSACTION	0 row(s) affected	0.000 sec

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	30000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	30000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

✓	46	10:55:09	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.000 sec / 0.000 sec
✓	47	10:55:26	COMMIT	0 row(s) affected	0.000 sec
✓	48	10:55:31	SET SESSION TRANSACTION ISOLATION L...	0 row(s) affected	0.000 sec
✓	49	10:55:34	START TRANSACTION	0 row(s) affected	0.000 sec
✓	50	10:55:36	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.015 sec / 0.000 sec

Program:

```
171 ● COMMIT;
172 ● SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE;
173 ● START TRANSACTION;
174 ● SELECT *
175 FROM Account
176 WHERE Account_No=10001;
177 ● COMMIT;
178 ● SELECT @@SESSION.transaction_isolation;
179 ● SELECT @@GLOBAL.transaction_isolation;
180 ● START TRANSACTION;
181 ● UPDATE Account
182 SET Balance=Balance+5000
183 WHERE Account_No=10001;
184 ● COMMIT;
185 ● START TRANSACTION;
186 ● UPDATE Account
187 SET Balance=Balance-5000
188 WHERE Account_No=10001;
189 ● COMMIT;
190 ● START TRANSACTION;
191 ● UPDATE Account
192 SET Balance=Balance-10000
193 WHERE Account_No=10001;
194 ● ROLLBACK;
195 ● START TRANSACTION;
196 ● UPDATE Account
197 SET Balance=Balance-10000
198 WHERE Account_No=10001;
```

Output:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	30000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Account 15 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 49	10:55:34	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 50	10:55:36	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.015 sec / 0.000 sec
✓ 51	10:57:31	COMMIT	0 row(s) affected	0.000 sec
✓ 52	10:57:33	SET SESSION TRANSACTION ISOLATION L...	0 row(s) affected	0.000 sec
✓ 53	10:57:34	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 54	10:57:37	SELECT * FROM Account WHERE Account_N...	1 row(s) returned	0.000 sec / 0.000 sec

@@SESSION.transaction_isolation

▶ SERIALIZABLE

✓ 55	10:58:09	COMMIT	0 row(s) affected	0.000 sec
✓ 56	10:58:11	SELECT @@SESSION.transaction_isolation LI...	1 row(s) returned	0.016 sec / 0.000 sec
✓ 57	10:58:25	SELECT @@GLOBAL.transaction_isolation LI...	1 row(s) returned	0.015 sec / 0.000 sec
✓ 58	10:58:28	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 59	10:58:32	UPDATE Account SET Balance=Balance+500...	1 row(s) affected Rows matched: 1 Changed: 1...	0.015 sec
✓ 60	10:58:38	COMMIT	0 row(s) affected	0.000 sec

✓ 63	10:59:26	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 64	10:59:28	UPDATE Account SET Balance=Balance-5000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓ 65	10:59:31	COMMIT	0 row(s) affected	0.000 sec
✓ 66	10:59:33	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 67	10:59:38	UPDATE Account SET Balance=Balance-1000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓ 68	10:59:42	ROLLBACK	0 row(s) affected	0.000 sec

✓ 69	10:59:44	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 70	11:00:07	UPDATE Account SET Balance=Balance-1000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec

Program:

```

200 ● SAVEPOINT AfterDebit;
201 ● UPDATE Account
202     SET Balance=Balance+10000
203     WHERE Account_No=10002;
204 ● COMMIT;
205 ● START TRANSACTION;
206 ● UPDATE Account
207     SET Balance=Balance+10000
208     WHERE Account_No=10001;
209 ● SAVEPOINT Deposit1;
210 ● UPDATE Account
211     SET Balance=Balance-3000
212     WHERE Account_No=10002;
213 ● ROLLBACK TO SAVEPOINT Deposit1;
214 ● COMMIT;
215 ● START TRANSACTION;
216 ● UPDATE Account
217     SET Balance=Balance-5000
218     WHERE Account_No=10001;
219 ● SAVEPOINT TestPoint;
220 ● UPDATE Account
221     SET Balance=Balance+5000
222     WHERE Account_No=10002;
223 ● ROLLBACK TO SAVEPOINT TestPoint;
224 ● RELEASE SAVEPOINT TestPoint;
225 ● COMMIT;

```

Output:

✓	71	11:01:50	SAVEPOINT AfterDebit	0 row(s) affected	0.015 sec
✓	72	11:01:52	UPDATE Account SET Balance=Balance+100...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	73	11:01:54	COMMIT	0 row(s) affected	0.016 sec
✓	74	11:01:56	START TRANSACTION	0 row(s) affected	0.000 sec
✓	75	11:01:59	UPDATE Account SET Balance=Balance+100...	1 row(s) affected Rows matched: 1 Changed: 1...	0.016 sec
✓	76	11:02:02	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
✓	77	11:02:18	UPDATE Account SET Balance=Balance-3000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	78	11:02:22	ROLLBACK TO SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
✓	79	11:02:24	COMMIT	0 row(s) affected	0.000 sec
✓	80	11:02:25	START TRANSACTION	0 row(s) affected	0.000 sec
✓	81	11:02:27	UPDATE Account SET Balance=Balance-5000...	1 row(s) affected Rows matched: 1 Changed: 1...	0.000 sec
✓	82	11:02:29	SAVEPOINT TestPoint	0 row(s) affected	0.000 sec

✓	83	11:02:48	UPDATE Account SET Balance=Balance+500...	1 row(s) affected Rows matched: 1 Changed: 1...	0.016 sec
✓	84	11:02:53	ROLLBACK TO SAVEPOINT TestPoint	0 row(s) affected	0.000 sec
✓	85	11:02:54	RELEASE SAVEPOINT TestPoint	0 row(s) affected	0.000 sec
✓	86	11:02:56	COMMIT	0 row(s) affected	0.016 sec